

NILESH SUBHASH CHILKA

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I'm **AI/ML Engineer** focused on building real-world AI products, especially in Generative AI, Machine Learning, and Computer Vision. I've worked with global enterprises to design production-grade AI systems, from RAG-based copilots to multi-agent platforms and real-time vision applications. My goal is to build AI solutions that are not just technically strong but also practical, scalable, and valuable for businesses and users.

TECHNICAL SKILLS

Programming Languages: Python

Databases: SQL Server, MySQL, MongoDB, Qdrant, Pinecone

Tools: Pandas, Numpy, Scikit-learn, Keras, Tensorflow, Pytorch, NLTK, Matplotlib, Seaborn, Flask, FastAPI, OpenCV, Selenium, Pyspark, LangChain, LlamaIndex, LangGraph, CrewAI, Google ADK, Docker, Kubernetes, Power BI, Terraform

Hardware: Raspberry PI, Arducam, SW-420 Vibrational Sensor, Load Cell Weight Sensor

Cloud Platforms: Microsoft Azure, Amazon Web Services (AWS), Google Cloud Platform (GCP)

WORK EXPERIENCE

HTC Global Services

Nov 2024 – Present

Senior Machine Learning Engineer - StateFarm (Client)

- Designed **Multi-Agent Architecture** using **LangGraph** to generate Test Scripts & Stored to **S3 Bucket**.
- Implemented **Nano Agents** to improve accuracy of browser-use Agent for Test Automation.
- Developed and exposed as **FastAPI** endpoint and **deployed on ROSA** for Version One Testing.
- Implemented **Prompt Injection** and **Content Moderation** in Responsible AI for Open Source Models.

Tech: LangGraph, Browser-use, LangChain, FastAPI, Claude Sonnet 4.6, S3 Bucket, DynamoDB, SNS Topic, SQS Queue, ROSA

Quantiphi Analytics Solutions

Nov 2022 – Nov 2024

Senior Machine Learning Engineer - Maersk (Client)

HR Copilot Gen AI

- Developed** LLM driven Retrieval Augmented Generation (RAG Fusion) using **LangChain, Llama Index**.
- Automated** Document Ingestion Pipeline for Indexing into Qdrant Vector DB using CronJob.
- Enhanced** API response time by **30%** by building FastAPI APIs and able retrieve the data parallelly.
- Enabled** real-time monitoring by **Power BI** Dashboard for metrics like Usage Cost, Logins, Feedback, etc.

Tech: Llama Index, FastAPI, Qdrant DB, Azure OpenAI, Blob Storage, CosmosDB, Docker, Kubernetes, CI/CD, Power BI.

Ernst & Young LLP.

Dec 2021 – Nov 2022

Machine Learning Engineer - Anheuser-Busch InBev (Client)

Smart OTC

- Architected** a scalable, end-to-end pipeline using ADF, Function App, and Container Instances.
- Automated** container management with Function App for deployment, status checks, and log storage
- Developed** Logic App for automated email triggers and file sharing based on container status.
- Designed** robust backend APIs for the web application using **Flask**.

Tech: Azure Data Factory, Flask, Container Instances, Function App, Logic App, Blob Storage, SQL Server.

Kaglorsys Technologies Private Limited.

Feb 2021 – Nov 2021

Data Scientist - Skylar Capital Management LP (Client)

Status Monitoring of Trains:

- Led** from requirements gathering from client to till Model deployment.
- Utilized** Selenium for data acquisition from web application and trained **YOLOv5** model using **PyTorch**.
- Developed** Flask based WebApp for live predictions and integrated **SQL Server** for data persistence.
- Implemented** automated alerts for status changes, driving operational efficiency.

Tech: OpenCV, YOLOv5, PyTorch, Pillow, Object Detection, Deep Learning, Flask, Selenium, SQL Server.

Facial Analysis viz. Gender, Age, Emotion & Active Speaker Detection

- Used Age, Gender & Emotion classification models for facial analysis.
- Trained **XGBoost** on time series distance between lips for **Active Speaker Detection**.
- Tracked each face to record **Gender, Age & Emotion** information in **S3 Bucket**.
- Visualized stored records using plots, exposed it as an API using Flask & developed **POC**.

Tech: OpenCV, RetinaNet, TensorFlow, Pillow, Object Detection, Deep Learning, XGBoost, S3 Bucket, Flask.

Real-Time Bike Detection, Tracking, Counting & Speed Estimation: [Demo Link](#)

- **Collected** road recordings & annotated data to train YOLOv5 model using **PyTorch**, achieving **0.85 mAP**.
- **Developed** a Flask WebApp to process live IP camera feeds for real-time bike detection.
- **Implemented** data storage for metrics such as entry and exit counts, enabling detailed analysis (**POC**).

Tech: OpenCV, YOLOv5, PyTorch, Pillow, Object Detection, Deep Learning, Flask, MySQL

PERSONAL PROJECTS

MagicMirror – AI Photo Clicker: [Github Link](#)

Sep 2025 – Sep 2025

- Implemented “Wave” & “Like” gesture recognition using Mediapipe to Click & Download Photo respectively.
- Used Nano Banana for Enhancing overall image of the guests and Pose Enhancement.
- Built QR Code to scan & Download Photos of Guests from FastAPI & deployed on Raspberry PI 4.

Tech: OpenCV, Mediapipe, Nano Banana, Pillow, FastAPI, Raspberry PI 4, ArduCam, Two Way Mirror

Spark Cricket – Cricket Simulation with Auto Scoring: [Github Link](#)

Aug 2025 – Aug 2025

- Collected the bowling action videos and used as bowling the balls from Projector.
- Used OpenCV to play the simulation and to overlay the Scoreboard, Fours, Sixes and Out.
- Used SW-420 Vibration sensor to detect the ball hit and count the score realtime based on location.
- Added commentary audio based on Fours, Sixes and Out using PyGame to make Engaging (POC)

Tech: OpenCV, SW-420 Vibration sensor, Pillow, Raspberry PI 4, ArduCam, PyGame

Analysis of India Stocks: [Demo Link](#)

Oct 2024 – Oct 2024

- Worked on Selenium to Navigate through screener.com and recorded video using pyautogui.
- Used computer vision techniques to identify key metrics like P/E, ROCE, Current Ratio, Reserves, etc
- Calculated a score based on identified key metrics as Fundamental Analysis score & stored in MySQL.

Tech: Python, OpenCV, Selenium, Pyautogui, Pillow, EasyOCR, Smtplib, MySQL, Moviepy.

FitBox – AI Fitness Trainer: [Demo Link](#)

Aug 2022 – Nov 2022

- Collected data for various exercises and poses like Pushups, Plank, Situps, Squats, Chair Pose, etc
- Used MediaPipe Python Library for Landmarks Prediction of the body.
- Based on X, Y Coordinates and Angles, classified the pose and action of the body and exposed as WebRTC
- Built Flutter based APP for Frontend and used Firebase for storing data and integrated Stripe for Payments.

Tech: OpenCV, MediaPipe, Pillow, WebRTC, gTTS, Stripe, Flutter, Android

Computer Vision for Blind Person: [Github Link](#) [Demo Link](#)

Dec 2020 – Jan 2021

- Used pre-trained **YOLOv3** model trained on COCO Dataset for Object Detection.
- Generated the speech such as 2 persons at center right, 2 glasses at center, 1 chair at bottom left.
- Deployed in App Engine in **Google Cloud Platform (GCP)** using **Flask**.

Tech: Python, OpenCV, YOLOv3, Object Detection, Pillow, gTTS, GCP, App Engine, Flask, TensorFlow.

Real-Time Face Mask Detection: [Github Link](#)

Oct 2020 – Oct 2020

- Collected the data from Google, Kaggle and trained **VGG16** model for classifying the face.
- Achieved Face Mask Detection **up to 20 meters** distance with CCTV camera.

Tech: Python, OpenCV, VGG16, Object Detection, Pillow, Keras, Fdet, PyTorch, TensorFlow

Signature Verification: [Github Link](#)**Aug 2020 – Sep 2020**

- Collected the data from Kaggle and pre-processed the data into Anchor, Positive, Negative.
- Trained **Siamese Network** using triplet-loss by using **Keras**.
- Deployed in **Heroku** using **Flask** which accepts two signatures for verification.

*Tech: Python, OpenCV, Siamese Network, Heroku, Pillow, Keras, Flask, TensorFlow, Numpy***Aadhaar Card Details Extractor:** [Github Link](#)**Jul 2020 – Jul 2020**

- Collected Aadhaar Card images from Google and trained **VGG16** CNN model for classifying the given image.
- Extracted the details from Aadhaar Card using **EasyOCR** and face by using Haarcascade Classifier.
- Stored all the details in **MySQL** Database.

*Tech: Python, OpenCV, MySQL, Pillow, Keras, EasyOCR, TensorFlow, VGG16***Sentiment Analysis of Covid-19 Tweets:** [Github Link](#)**Jun 2020 – Jul 2020**

- Downloaded the dataset (sentiment140) from Kaggle and preprocessed using **NLTK**.
- Trained **LSTM** model using **Keras** and analyzed the sentiments of live tweets by plotting.

*Tech: Python, NLTK, LSTM, Keras, TensorFlow, Twitter API***EDUCATION****2017 – 2021**

Walchand Institute of Technology.

*B.E. in Electronics & Telecommunication***CERTIFICATIONS****Microsoft Certified: Azure Data Scientist Associate** - Microsoft**Applied Data Science with Python Specialization** - Coursera

- Introduction to Data Science in Python.
- Applied Plotting, Charting and Data Representation in Python.
- Applied Machine Learning in Python.
- Applied Text Mining in Python.
- Applied Social Network Analysis in Python.

Deep Learning Specialization - Coursera

- Neural Networks and Deep Learning.
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization
- Structuring Machine Learning Projects.
- Convolutional Neural Networks.
- Sequence Models.

Blogs

- Practical Guide to OpenCV - [Link](#)

Published Packages

- AutoEnsembler - [Link](#)